

Digital Electronics Study Notes (EC204)

Academic Insight - Public Sample Revision Summary

Module 1: Boolean Algebra & Karnaugh Map Minimization

- De Morgan Laws: $(A.B)' = A' + B'$, $(A + B)' = A' . B'$
- Consensus Theorem: $A B + A' C + B C = A B + A' C$
- K-Map uses Gray code ordering (00, 01, 11, 10) to group power-of-2 minterms.

Module 2: Synchronous Sequential Circuits & Timing

- Flip-Flop equations: D: $Q(t+1) = D$, JK: $Q(t+1) = J Q' + K' Q$, T: $Q(t+1) = T \text{ xor } Q$
- Max clock frequency: $T_{\text{clk}} \geq t_{\text{cq}} + t_{\text{logic,max}} + t_{\text{su}}$
- Note: Sample educational revision sheet provided by Academic Insight.